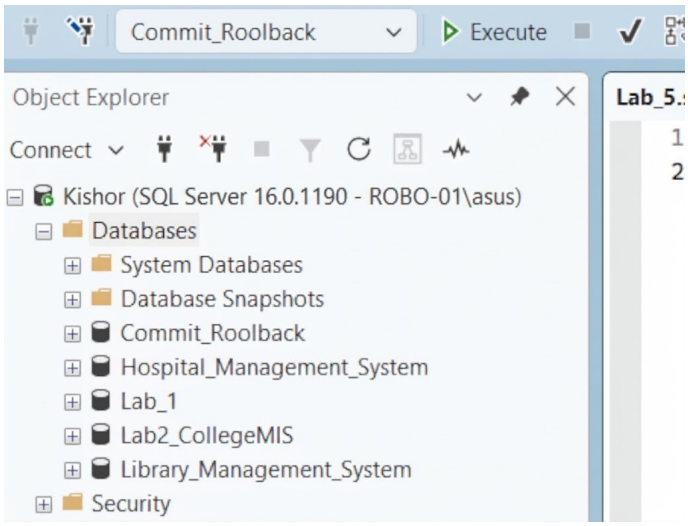
Database: Commit_Rollback

Q1: Create the database LibraryMS and connect to the database.

Query:

```
CREATE DATABASE Commit_Rollback;
USE Commit_Rollback;
```

Output:

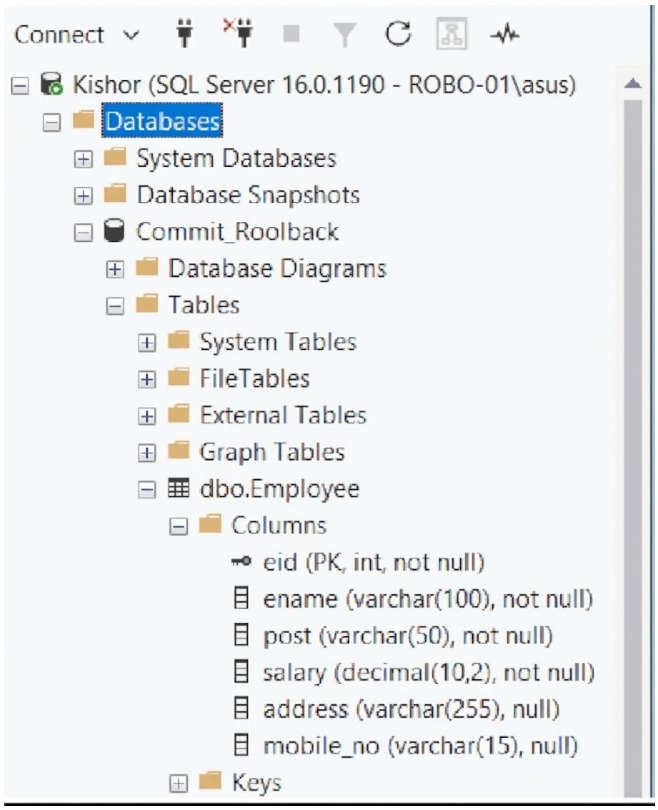


Q2: Create a table named Employee.

Query:

```
CREATE TABLE Employee (
    eid INT PRIMARY KEY IDENTITY(1,1),
    ename VARCHAR(100) NOT NULL,
    post VARCHAR(50) NOT NULL,
    salary DECIMAL(10, 2) NOT NULL,
    address VARCHAR(255) NULL,
    mobile_no VARCHAR(15) NULL
);
```

Output:



Q3: Insert any 10 marks into the table.

<u>Query:</u>

```
INSERT INTO Employee (ename, post, salary, address, mobile_no)
VALUES
    ('Ramesh Poudel', 'Software Developer', 68000.00, 'Kathmandu, Nepal', '9812345670'),
    ('Sunita Tamang', 'Project Coordinator', 95000.00, 'Lalitpur, Nepal', '9823456781'),
    ('Aayush KC', 'Graphic Designer', 58000.00, 'Bhaktapur, Nepal', '9834567892'),
    ('Bimala Gurung', 'HR Manager', 72000.00, 'Pokhara, Nepal', '9845678903'),
    ('Prakash Magar', 'Cloud Engineer', 90000.00, 'Biratnagar, Nepal', '9856789014'),
    ('Sabina Shahi', 'Test Engineer', 62000.00, 'Kathmandu, Nepal', '9867890125'),
    ('Nabin Shrestha', 'Database Administrator', 82000.00, 'Chitwan, Nepal', '9878901236'),
    ('Karuna Lama', 'Network Administrator', 76000.00, 'Butwal, Nepal', '9889012347'),
```

('Dipesh Basnet', 'Full Stack Developer', 85000.00, 'Lalitpur, Nepal', '9890123458'),

('Rojina Rai', 'Mobile App Developer', 74000.00, 'Dharan, Nepal', '9801234569');

SELECT * FROM Employee;

Output:

Results | Messages

	eid	ename	post	salary	address	mobile_no
1	1	Ramesh Poudel	Software Developer	68000.00	Kathmandu, Nepal	9812345670
2	2	Sunita Tamang	Project Coordinator	95000.00	Lalitpur, Nepal	9823456781
3	3	Aayush KC	Graphic Designer	58000.00	Bhaktapur, Nepal	9834567892
4	4	Bimala Gurung	HR Manager	72000.00	Pokhara, Nepal	9845678903
5	5	Prakash Magar	Cloud Engineer	90000.00	Biratnagar, Nepal	9856789014
6	6	Sabina Shahi	Test Engineer	62000.00	Kathmandu, Nepal	9867890125
7	7	Nabin Shrestha	Database Administrator	82000.00	Chitwan, Nepal	9878901236
8	8	Karuna Lama	Network Administrator	76000.00	Butwal, Nepal	9889012347
9	9	Dipesh Basnet	Full Stack Developer	85000.00	Lalitpur, Nepal	9890123458
10	10	Rojina Rai	Mobile App Developer	74000.00	Dharan, Nepal	9801234569

Q4: Change the salary of the employees by 1.10.

Query:

```
begin transaction;
UPDATE Employee
SET salary=salary*1.10;
```

Output:

Results | Messages

	eid	ename	post	salary	address	mobile_no
1	1	Ramesh Poudel	Software Developer	74800.00	Kathmandu, Nepal	9812345670
2	2	Sunita Tamang	Project Coordinator	104500.00	Lalitpur, Nepal	9823456781
3	3	Aayush KC	Graphic Designer	63800.00	Bhaktapur, Nepal	9834567892
4	4	Bimala Gurung	HR Manager	79200.00	Pokhara, Nepal	9845678903
5	5	Prakash Magar	Cloud Engineer	99000.00	Biratnagar, Nepal	9856789014
6	6	Sabina Shahi	Test Engineer	68200.00	Kathmandu, Nepal	9867890125
7	7	Nabin Shrestha	Database Administrator	90200.00	Chitwan, Nepal	9878901236
8	8	Karuna Lama	Network Administrator	83600.00	Butwal, Nepal	9889012347
9	9	Dipesh Basnet	Full Stack Developer	93500.00	Lalitpur, Nepal	9890123458
10	10	Rojina Rai	Mobile App Developer	81400.00	Dharan, Nepal	9801234569

Q5: Rollback to the previous salary.

Query:

```
rollback
SELECT * FROM Employee
```

Output:

Results | Messages

	eid	ename	post	salary	address	mobile_no
1	1	Ramesh Poudel	Software Developer	68000.00	Kathmandu, Nepal	9812345670
2	2	Sunita Tamang	Project Coordinator	95000.00	Lalitpur, Nepal	9823456781
3	3	Aayush KC	Graphic Designer	58000.00	Bhaktapur, Nepal	9834567892
4	4	Bimala Gurung	HR Manager	72000.00	Pokhara, Nepal	9845678903
5	5	Prakash Magar	Cloud Engineer	90000.00	Biratnagar, Nepal	9856789014
6	6	Sabina Shahi	Test Engineer	62000.00	Kathmandu, Nepal	9867890125
7	7	Nabin Shrestha	Database Administrator	82000.00	Chitwan, Nepal	9878901236
8	8	Karuna Lama	Network Administrator	76000.00	Butwal, Nepal	9889012347
9	9	Dipesh Basnet	Full Stack Developer	85000.00	Lalitpur, Nepal	9890123458
10	10	Rojina Rai	Mobile App Developer	74000.00	Dharan, Nepal	9801234569

Q6: Commit to it.

Query:

```
begin transaction;
commit
SELECT * FROM Employee
```

Output:

Results | Messages

	eid	ename	post	salary	address	mobile_no
1	1	Ramesh Poudel	Software Developer	68000.00	Kathmandu, Nepal	9812345670
2	2	Sunita Tamang	Project Coordinator	95000.00	Lalitpur, Nepal	9823456781
3	3	Aayush KC	Graphic Designer	58000.00	Bhaktapur, Nepal	9834567892
4	4	Bimala Gurung	HR Manager	72000.00	Pokhara, Nepal	9845678903
5	5	Prakash Magar	Cloud Engineer	90000.00	Biratnagar, Nepal	9856789014
6	6	Sabina Shahi	Test Engineer	62000.00	Kathmandu, Nepal	9867890125
7	7	Nabin Shrestha	Database Administrator	82000.00	Chitwan, Nepal	9878901236
8	8	Karuna Lama	Network Administrator	76000.00	Butwal, Nepal	9889012347
9	9	Dipesh Basnet	Full Stack Developer	85000.00	Lalitpur, Nepal	9890123458
10	10	Rojina Rai	Mobile App Developer	74000.00	Dharan, Nepal	9801234569

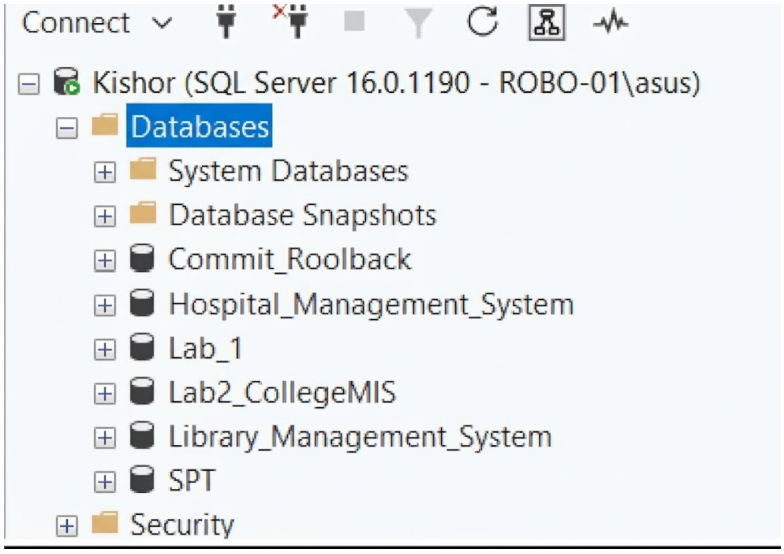
Database: SPT

Q1: Create the database SPT and connect to the database.

Query:

```
CREATE DATABASE SPT;
USE SPT;
```

Output:



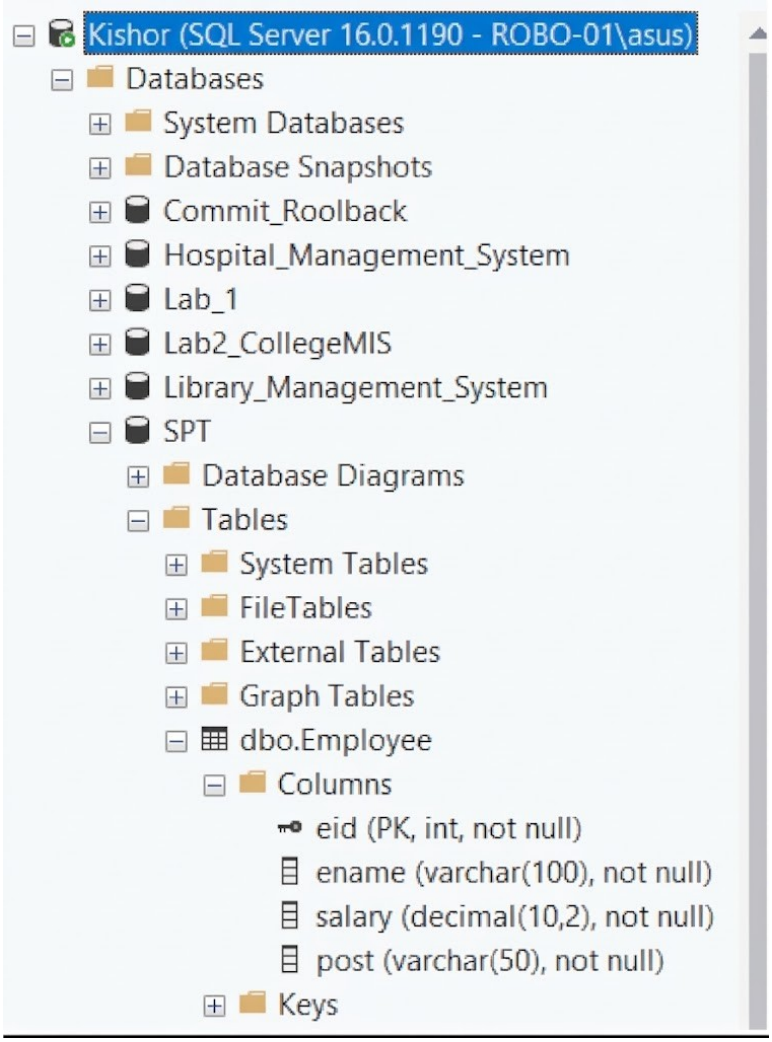
Q2: Create a table named Employee.

Query:

```
CREATE TABLE Employee (
   eid INT PRIMARY KEY IDENTITY(1,1),
   ename VARCHAR(100) NOT NULL,
   salary DECIMAL(10, 2) NOT NULL,
   post VARCHAR(50) NOT NULL

);
```

Output:



Q3: Insert any 10 marks into the table.

<u>Query:</u>

```
INSERT INTO Employee VALUES
('Kamal Poudel', 88000.00, 'Software Architect'),
('Rina Shrestha', 115000.00, 'Project Manager'),
('Bibek Adhikari', 52000.00, 'Software Developer'),
('Mina Gurung', 92000.00, 'Database Administrator'),
('Suman Karki', 65000.00, 'Software Engineer'),
('Roshan Thapa', 105000.00, 'DevOps Engineer'),
('Anu Bhandari', 42000.00, 'Intern'),
('Prabin Rai', 78000.00, 'System Analyst'),
('Sarita Lama', 125000.00, 'Department Head'),
('Nischal Tamang', 55000.00, 'Support Engineer');
```

```
SELECT * FROM Employee;
```

Output:

Results | Messages

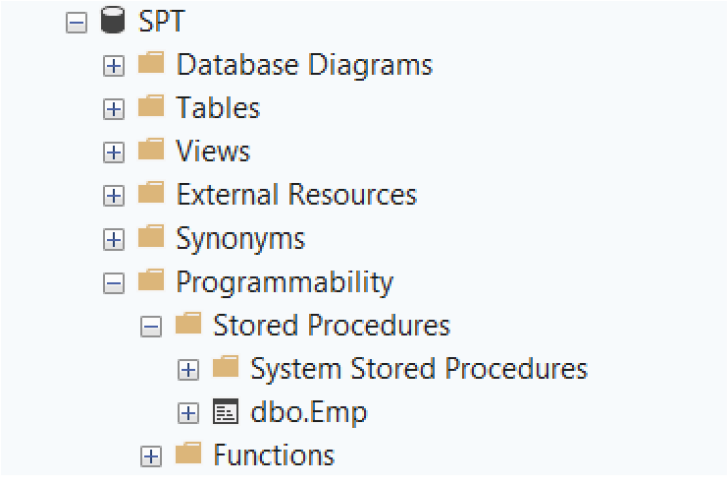
	eid	ename	salary	post
1	1	Kamal Poudel	88000.00	Software Architect
2	2	Rina Shrestha	115000.00	Project Manager
3	3	Bibek Adhikari	52000.00	Software Developer
4	4	Mina Gurung	92000.00	Database Administrator
5	5	Suman Karki	65000.00	Software Engineer
6	6	Roshan Thapa	105000.00	Senior Software Engineer
7	7	Anu Bhandari	42000.00	Intern
8	8	Prabin Rai	78000.00	System Analyst
9	9	Sarita Lama	125000.00	Department Head
10	10	Nischal Tamang	55000.00	Support Engineer

Q4: Create a stored procedure to display record for Senior Software Engineer.

Query:

```
CREATE PROCEDURE Emp
AS
BEGIN
   SELECT *
   FROM Employee
   WHERE post = 'Senior Software Engineer';
END;
```

Output:



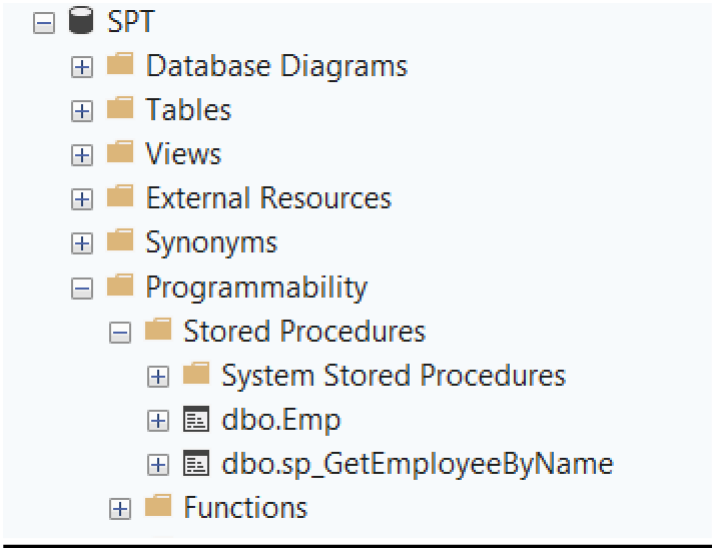
Q5: Create a stores procedure for input name parameter and display their details.

Query:

```
SELECT * FROM Employee CREATE PROCEDURE
sp_GetEmployeeByName
   @ename VARCHAR(100)
AS
BEGIN
   SELECT *
   FROM Employee
   WHERE ename = @ename;
END;
GO


EXEC sp_GetEmployeeByName @ename = Roshan Thapa';
```

Output:



Results | Messages

	eid	ename	salary	post
1	6	Roshan Thapa	105000.00	Senior Software Engineer

Q6: Create a trigger to alert a message after inserting data to the Employee table.

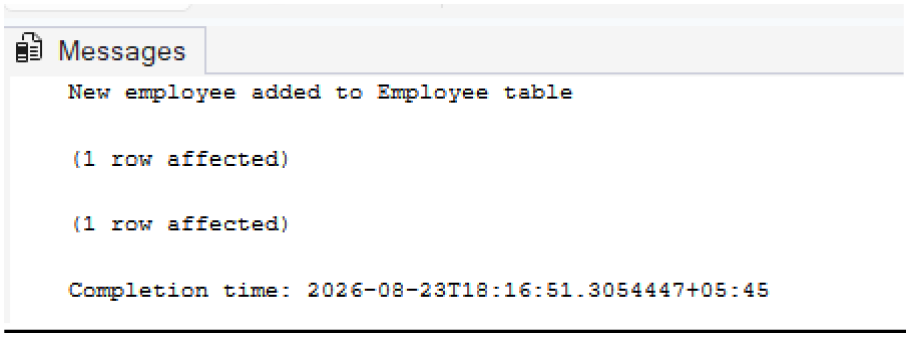
Query:

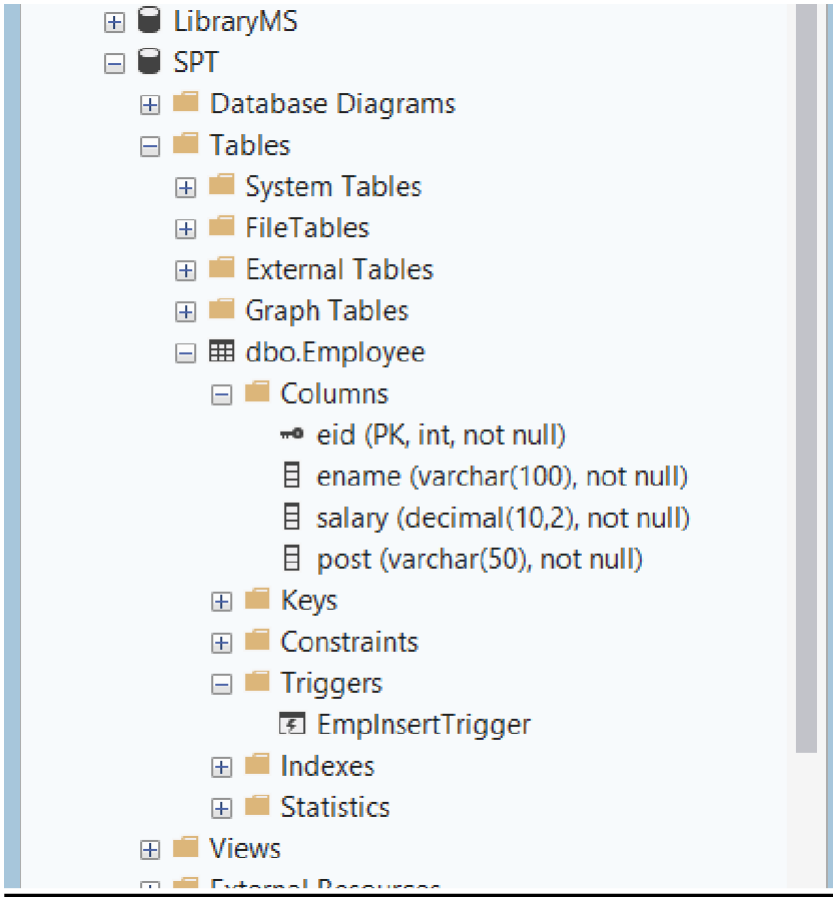
```
CREATE TRIGGER EmpInsertTrigger
```

```
ON Employee
After INSERT
AS
BEGIN
   PRINT 'New employee added to Employee table';
END;

INSERT INTO Employee (ename, salary, post) VALUES
('Rabi Bhatt', 105000.00, 'AI & Robotics Engineer');
```

Output:





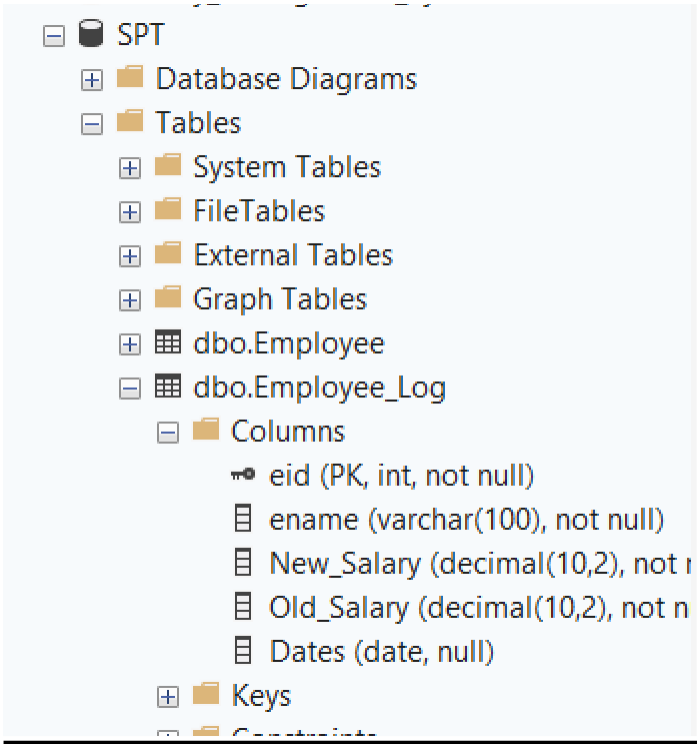
Q7: Create a table named EmployeeLog .

Query:

```
CREATE TABLE Employee_Log (
    eid INT PRIMARY KEY IDENTITY(1,1),
    ename VARCHAR(100) NOT NULL,
    New_Salary DECIMAL(10, 2) NOT NULL,
    Old_Salary DECIMAL(10, 2) NOT NULL,
    Dates DATE,

);
```

Output:



Q8: Create a trigger named EmployeeUpdateLogTrigger that stores historical data after updating data to the employee table.

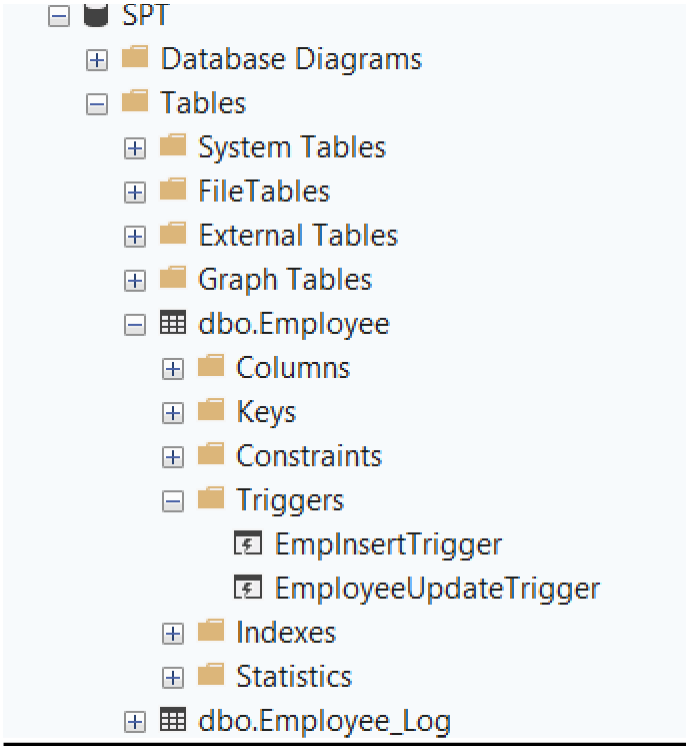
Query:

```
SET IDENTITY_INSERT Employee_Log ON;

CREATE TRIGGER EmployeeUpdateTrigger
ON Employee
AFTER UPDATE
AS
BEGIN
```

```
    INSERT INTO Employee_Log (eid, ename, New_Salary,
Old_Salary, Dates)
    SELECT
        d.eid,
        i.ename,
        i.salary,
        d.salary,
        CAST(GETDATE() AS DATE)
    FROM DELETED d
    JOIN INSERTED i ON d.eid = i.eid;
END;
GO
```

Output:



Q9: Update the data in the Employee Table making the salary of all 20000.

Query:

```
UPDATE  Employee
SET Salary = 20000;
GO
SELECT * FROM Employee;
SELECT * FROM Employee_Log;
```

Output:

Results | Messages

	eid	ename	salary	post
1	1	Kamal Poudel	20000.00	Software Architect
2	2	Rina Shrestha	20000.00	Project Manager
3	3	Bibek Adhikari	20000.00	Software Developer
4	4	Mina Gurung	20000.00	Database Administrator
5	5	Suman Karki	20000.00	Software Engineer
6	6	Roshan Thapa	20000.00	Senior Software Engineer
7	7	Anu Bhandari	20000.00	Intern
8	8	Prabin Rai	20000.00	System Analyst
9	9	Sarita Lama	20000.00	Department Head
10	10	Nischal Tamang	20000.00	Support Engineer
11	11	Rabi Bhatt	20000.00	AI & Robotics Engineer

	eid	ename	New_Salary	Old_Salary	Dates
1	1	Kamal Poudel	20000.00	88000.00	2026-08-23
2	2	Rina Shrestha	20000.00	115000.00	2026-08-23
3	3	Bibek Adhikari	20000.00	52000.00	2026-08-23
4	4	Mina Gurung	20000.00	92000.00	2026-08-23
5	5	Suman Karki	20000.00	65000.00	2026-08-23
6	6	Roshan Tha...	20000.00	105000.00	2026-08-23
7	7	Anu Bhandari	20000.00	42000.00	2026-08-23
8	8	Prabin Rai	20000.00	78000.00	2026-08-23
9	9	Sarita Lama	20000.00	125000.00	2026-08-23
10	10	Nischal Tam...	20000.00	55000.00	2026-08-23
11	11	Rabi Bhatt	20000.00	105000.00	2026-08-23